SportSathi: An AI-Based Fitness Screening and Sports Talent Guidance Platform for Schools

Project Report — Prototype Stage, for UCS503P
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1 Purpose of this stage

This report captures the state of SportSathi midway between the initial proposal and the final report. Its purpose is narrower than the final report: to show that the riskiest technical assumption in the proposal — that an explainable classifier can be trained honestly on available student fitness data — has been tested, and to record what was learned before the remaining workflow (coach approval, dashboards, reporting) was built out.

2 What was prototyped

- A public student physical-education performance dataset (500 records, 20 attributes) was identified and adopted as the Tier 1 training/evaluation dataset, in place of collecting real school data, consistent with the proposal's privacy constraints.
- A first baseline classifier (Decision Tree) was trained and evaluated in isolation, ahead of any backend integration, exactly as the proposal's time-to-value heuristic specified.
- The assessment-entry form and BMI auto-calculation were scaffolded on the frontend.

3 Key finding: a leakage problem the proposal did not anticipate

The first baseline run included a derived `Overall_PE_Performance_Score` column as a model input and produced a suspicious 98% accuracy. Feature importance analysis showed the model relying almost entirely on that single derived column and on attendance, while every genuine physical fitness score (strength, endurance, flexibility, speed-agility) had zero importance — a clear sign the derived column already encoded the answer.

This was corrected by excluding all derived/summary columns (`Overall_PE_Performance_Score`, `Improvement_Rate`, `Final_Grade`) from the feature set. Retraining on only genuine, independently collected attributes produced a more credible 81–86% accuracy (Decision Tree: 86%, Random Forest: 81%).

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This finding is the main output of the prototype stage. It confirms the proposal's "validate in isolation before integration" heuristic was the right call: the problem was caught while the model was still being tested on its own, before it was wired into any user-facing workflow.

4 Secondary finding: attendance as a strong predictor

Class-wise average attendance showed a clean, monotonic separation across performance categories (High Performer: 94.2%, Average: 89.8%, Low: 84.7%), stronger than any single raw physical-fitness score at separating the classes. This was not anticipated in the proposal and is carried forward as a candidate insight to validate further in the final stage.

5 Known limitation carried forward

The dataset's target classes are heavily imbalanced (245 Low Performer, 243 Average Performer, 12 High Performer). Both the Decision Tree and Random Forest baselines failed to correctly predict any High Performer example. This is flagged here, before further work, as a limitation to be explicitly addressed (or explicitly documented as unresolved) in the final report, rather than discovered late.

6 Status against the proposal's initial deliverable list

Proposed initial deliverable	Status at prototype stage
Authentication and role-based access	Not yet started
Tier 1 assessment entry + BMI calc.	Frontend scaffolded
Trained, evaluated classifier	Baseline complete; leakage found and corrected
Coach approval/override workflow	Not yet started
CI: build, tests, linting	Not yet started
CD to staging	Not yet started

7 Revised plan for the final stage

- Integrate the corrected classifier into the backend, behind a coach-approval gate, before adding any further dashboard features.
- Apply class-imbalance handling (SMOTE, class weighting) to the classifier and report per-class metrics, not just overall accuracy, in the final evaluation.
- Prioritise the coach-approval workflow and student/parent visibility gating over CI/CD if a time trade-off becomes necessary, since the approval gate is the project's core safety property.
- Treat sport-suitability recommendation logic and the full relational database migration as stretch goals for the final stage, to be reported honestly as complete or incomplete rather than assumed.

8 Summary

The prototype stage validated the proposal's central technical risk — that an explainable, honestly-evaluated fitness classifier is achievable on available data — and, in doing so, surfaced a data-leakage problem the proposal had not anticipated. Catching this early, during isolated model testing rather than after integration, is treated here as evidence that the proposal's time-to-value approach is working as intended. The remaining workflow components (coach approval, dashboards, CI/CD) are carried into the final stage as the primary remaining scope.